MILITARY SPECIFICATION

WELDING SET, ARC: AC/DC, MOBILE MULTIPROCESS, DIESEL ENGINE DRIVEN

This specification is approved for use the by Marine Corps, Department of Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type diesel engine driven, trailer mounted, rectified alternating current (ac) or direct current (dc), constant current, constant voltage mobile welding set mounted on a Government furnished M353 trailer (see Figure 1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

VV-F-800 - Fuel Oil, Diesel

ZZ-H-461 - Hose and Hose Assembly, Rubber, Gas (Acetylene-Hydrogen, Air and Oxygen).

MILITARY

MIL-V-173 - Varnish, Moisture and Fungus Resistant (For Treatment of Communication, Electronic and Associated Equipment)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commandant of the Marine Corps, Headquarters Marine Corps, Code LMA-1, Washington, D. C. 20380 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:42Z
Check the source to verify that this is the current version before use.

MIL-T-704 - Treatment and Painting of Materiel

MIL-L-2104 - Lubricating Oil, Internal Combustion Engine, Tactical Service

MIL-L-2105 - Lubricating Oil, Gear, Multi-Purpose

MIL-T-5624 - Turbine Fuel, Aviation, Grade JP-4 and JP-5

MIL-G-10924 - Grease, Automotive and Artillery

MIL-R-19180 - Regulator, Pressure, Compressed Gas

MIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging of

MIL-L-46167 - Lubricating Oil, Internal Combustion Engine, Arctic

MIL-E-52798 - Enamel, Alkyd, Camouflage

MIL-C-62122 - Cable Assembly, Slave, Electrical, Connectors, Plug and Receptacle

STANDARDS

FEDERAL

FED-STD-595 - Color

MILITARY

MIL-STD-129 - Marking for Shipment and Storage

MIL-STD-130 - Identification Marking of U.S. Military Property

MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference

MIL-STD-462 - Electromagnetic Interferences Characteristics, Measurement of

MIL-STD-642 - Identification Marking of Combat and Tactical Transport Vehicles

MS35000-3 - Battery, Storage, Lead-Acid, Waterproof

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:42Z
Check the source to verify that this is the current version before use.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

ENVIROMENTAL PROTECTION AGENCY (EPA)

40 CFR, Part 85 - Control of Air Pollution from New Motor Vehicles and new Motor Vehicle Engines.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

DEPARTMENT OF LABOR (DOL)

29 CFR 1926 - Occupational Safety and Health Administration (OSHA) Standards for Construction Jobs.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C. 20402. When indicated, reprints of certain regulations may be obtained from the federal agency responsible for issuance thereof.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herin, the text of this specification shall take precedence.

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION INC (NMFTA)

National Motor Freight Classification Rules

(Application for copies should be addressed to the American Trucking Association, Inc., Traffic Dept., 1616 P Street, N.W., Washington, D. C. 20036.)

UNIFORM CLASSIFICATION COMMITTEE (UCC)

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:42Z
Check the source to verify that this is the current version before use.

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA) STANDARDS

EW-1 Electric Arc Welding Apparatus

WD-1 Wiring Devices

(Application for copies should be addressd to the National Electrical Manufacturers Association, 2101 L Street, NW., Washington, D.C. 20037.)

AMERICAN WELDING SOCIETY (AWS)

Welding Handbook

(Application for copies should be addressed to American Welding Society, 2501 N.W. 7th Street, Miami, Florida 33125.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

J 429 Mechanical and Material Requirements for External Threaded Fasteners

J 816b Engine Test Code

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., Two Pennsylvania Plaza, New York, New York 10001.)

FACTORY MUTUAL RESEARCH

OE5A0.AF (7813) Fuel System

(Application for copies should be addresed to Factory Mutual Research, 1151 Boston-Providence Turnpike, Norwood, Massachusetts 02062.)

Technical Society and Technical Association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using federal agencies.

3. REQUIREMENTS

3.1 First article. One complete mobile welding set with all components, subsystem assemblies and accessories as specified herein, conforming to all requirements of this specification, shall be submitted for first article inspection and tests. Approval of the first article shall not relieve the supplier of the responsibility of furnishing equipment in accordance with this specification. The first article shall remain at the manufacturer's plant as a sample and shall be delivered as the last unit of the initial contract. The approved sample shall be used as the standard for all production items manufactured to fulfill the terms of the contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:42Z
Check the source to verify that this is the current version before use.

3.2 Material. All material used in the manufacture of the welding set shall be new and of the quality necessary to produce welding sets to meet the requirements described herein. When material for a part is not definitely specified, the material shall be of the type normally used by the manufacturer. Materials shall be free from defects or imperfections that would contribute to failure of the unit or would materially reduce the usability and performance of the unit.

3.2.1 Fungus and moisture resistance. The electrical circuitry, including all components and connections, shall be protected from the effects of moisture and fungus growth by an overall treatment conforming to MIL-V-173 except as follows:

(a) Components or circuit elements that are inherently fungus and moisture resistant, or which are sealed hermetically, need not be treated.

(b) Components or circuit elements whose functions would be affected adversely by varnish coating shall not be treated.

3.3 Design. The welding set shall be a multipurpose, ac rectified or dc generator type driven by a liquid cooled diesel engine capable of providing constant current, constant voltage weld power and a minimum of 2 kilovoltampere (kVA) of 115 volt (V) ac auxillary power for the operation of power tools. The welding set shall be the current commercial production model of a standard product, and except for product improvements which have been incorporated on the commercial production units, shall have been marketed for at least 1 year prior to the opening of technical proposals or bids. The welding set shall be rugged in design and suitable for outdoor operations. The design shall include applicable requirements of NEMA EW-1 and the requirements specified herein. The welding set shall be mounted on a Government furnished trailer (see Figure 1).

3.3.1 Components. The welding set shall consist of, but not be limited to, the following components: engine, fuel system, generator, generator exciter, wiring, controls, auxiliary power supply, and enclosure.

3.3.1.1 Engine. The engine shall be a liquid cooled, diesel engine with sufficient power and speed rating to operate the welding generator and accessories at maximum rated capacity. The engine shall be mounted on rubber isolators to reduce noise and vibration. The engine shall be capable of operating and meeting the performance specified herein using diesel fuel conforming to VV-F-800 and JP-5 turbine fuel conforming to MIL-T-5624. The engine shall rotate at a maximum speed not to exceed 1950 revolutions per minute.

3.3.1.1.1 Starting system. The welding set shall be furnished with a 24-V electric starting system with sufficient batteries to crank the engine under all conditions specified herein. Batteries shall be

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:42Z
Check the source to verify that this is the current version before use.

6TN in accordance with MS35000-3. The starting system shall incorporate a slave receptacle conforming to MIL-C-62122 for starting the engine from auxiliary sources. The receptacle shall be located in a position accessible to a person standing on the groundn The engine starting motor shall be activated by either a switch or button. The engine electrical system shall not require a key to be activated.

3.3.1.1.2 Engine lubrication system. The engine lubrication system shall provide lubrication for all moving engine parts for continuous operation. The lubrication system shall be a full pressure type system. The welding set shall have oil drain provisions for draining the engine crankcase without the removal of any components. A bayonet gage or dip stick which is calibrated for determining crankcase oil level shall be provided. The welding set shall be equipped with a low oil pressure shutdown device to stop the engine in the event of inadequate oil pressure.

3.3.1.1.3 Battery charging system. The welding set shall have a battery charging system with sufficient capacity to keep the batteries charged for starting under all conditions specified herein.

3.3.1.1.4 Governor. The welding set shall have a constant speed, mechanical type governor which shall maintain the speed of the engine within 5 percent of throttle setting at rated operating speed. The governor shall be belt or gear driven.

3.3.1.1.5 Cooling system. the welding set shall have a liquid cooling system with thermostatic control. The radiator and fan shall be of adequate size and capacity so that the welding set can operate in ambient temperatures ranging from - 25°F to + 125°F. The outside of the radiator shall be protected from damage by a radiator guard. The welding set shall have a high temperature shutdown device to stop the engine in the event of excessive temperature.

3.3.1.1.6 Air cleaner. The engine shall have a dry type filter, the capacity of which shall be in accordance with the engine manufacturer's recommendation. A visual warning device showing when service should be performed shall be furnished.

3.3.1.1.7 Exhaust. The engine exhaust system shall be under the hood or otherwise protected to preclude water entering the engine during high wind rain storms.

3.3.1.1.8 Starting aid. The engine shall be provided with a pressurized container, 20 ounce capacity, measured shot ether cold weather starting aid. The control for the starting aid shall be mounted on the instrument panel and shall be operable with a single mittened hand.

3.3.1.2 Fuel system. The welding set shall have a fuel system with primary and secondary fuel filters, fuel tank, fuel lines, fuel pump, fuel gage and hand primer pump.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:42Z
Check the source to verify that this is the current version before use.

3.3.1.2.1 Fuel tank. The fuel tank shall have a capacity to operate the welding set for 8 hours at rated duty cycle, full load. The fuel tank filler cap shall be an approved diesel service cap capable of accepting a 1-3/4-inch refueler nozzle. A drain plug shall be provided in the bottom of the fuel tank. The tank shall conform with Factory Mutual Research Standard OE5A0.AF (7813).

3.3.1.2.2 Fuel lines. Fuel lines shall be free of sharp bends or loops that might cause failure from fatigue, strain or vapor lock. The fuel lines shall be bracketed and fastened when necessary to prevent vibration.

3.3.1.2.3 Fuel gage. The fuel tank shall have a gage to register the quantity of fuel in the tank at all times. The scale on the gage shall be marked in quarter increments, such as 0, 1/4, 1/2, 3/4, F.

3.3.1.3 Welding generator. The welding generator shall be a compound wound dc generator or a rectified ac alternator, rated at 320 amperes (A) at 40 load V if rated for a 100-percent duty cycle or rated at 400 A at 40 load V rated at 60-percent duty cycle. The welding generator shall be capable of both constant current and constant voltage under the conditions specified in 3.8.1 and 3.8.2 and shall conform to the standards in NEMA EW-1. The rotor shall be mounted in a double-scaled, anti-friction bearing. The rotor shall be close coupled to the engine through a metal coupling to maintain alignment with the engine flywheel.

3.3.1.3.1 Generator output studs/posts. Four output studs/posts shall be provided (work stud, manual welding stud, semiautomatic stud, earth ground stud) and shall be of the stud and wing or hex nut type. The studs shall be not less than 1/2 inch in diameter and shall be recessed, protected or located to prevent accidental contact. All studs shall be specifically identified. A 12-inch length of 2/0 welding cable in accordance with 3.3.4.2 with one lug and one female quick disconnect shall be attached to each stud.

3.3.1.3.2 Electrical insulation. The class of electrical insulation system used in the generator and exciter shall be not less than Class B of NEMA EW-1.

3.3.1.4 Generator exciter. The exciter shall either be integral with the welding generator, or separate and mounted on the same shaft as the main welding generator. The exciter shall have fast voltage buildup and shall be designed to prevent polarity reversal. The commutator, brushes, and holders or slip rings shall be readily accessible for servicing and maintenance.

3.3.1.5 Wiring. The type and size of wire used shall have temperature and voltage rating and insulation in accordance with NEMA WD-1. Wiring subject to chafing or vibration shall be secured to prevent damage. The wire and terminals shall be color or number coded. A wiring

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:42Z
Check the source to verify that this is the current version before use.

diagram of the generator and engine shall be provided and fastened in a convenient location to the inside surface of the welding systems enclosure. The diagram shall have a transparent protective covering or coating.

3.3.1.6 Controls and instruments. The welding set shall be complete with all necessary controls and instruments to start, stop and operate the engine at required speed; to start, stop and operate the welding power units at all available outputs; and to indicate all necessary operating conditions of both. All controls and instruments shall be grouped together on a common panel, flush mounted, panel type, designed for protection against dust and moisture. The instruments shall be accurate within 3 percent of full scale reading. The controls shall be designed for easy operation with a gloved or mittened hand. Engine speed control shall include a manual or automatic throttle to bring engine speed to a slow idle. The instrument and control panel shall be fitted with, but not necessarily limited to, the following:

(a) Engine coolant temperature gage

(b) Engine oil pressure gage

(c) Ammeter - battery charging

(d) Hourmeter

(e) Voltmeter - output voltage

(f) Ammeter - welding current

(g) Variable voltage control rheostat

(h) Constant voltage control rheostat

(i) Constant current/constant voltage selection switch

(j) Control - welding current - course adjustment

(k) Control - welding current - fine adjustment

(l) Control - engine speed

(m) Emergency shutdown switch

(n) Auxiliary power receptacles

All controls, instruments, and other items requiring identification shall be permanently and legibly marked for identification, position and purpose.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:42Z
Check the source to verify that this is the current version before use.

3.3.1.7 Auxiliary power supply. The welding set shall be equipped with an auxiliary power supply. Two grounding type, 115 V duplex receptacles conforming to NEMA WD-1 shall be provided for lights and power tools. The circuit shall be overload protected by a circuit breakers. The auxiliary supply shall be rated at not less than 2 KVA at 115 V ac 60 hertz(Hz). The welding set shall also be equipped with two three wire, twist-lock receptacles for hook up of wire feeders.

3.3.1.8 Enclosure. The welding set shall be enclosed in a weather-proofed sheet metal enclosure. The enclosure shall have hinged or removable panels for access and operation. The enclosure shall be such that internal parts are protected from the weather even when the enclosure is opened in the operating position. Design and location of ventilation openings shall be such as to prevent water from being drawn into the enclosure by the air cooling system. Instruments and controls shall be protected by piano hinged drip-proof doors. The sheet metal of the enclosure shall be not less than 16 gage (0.0568 inch or 1.44 millimeters) in thickness.

3.3.2 Subassembly systems and components. The manufacturer shall equip each welding set with the following subassembly systems and components. All subassembly systems and components supplied shall be completely compatible with the welding set produced by the manufacturer.

3.3.2.1 Metal inert gas welder. This system shall consist of the wire feeder, cables, gun and wire spools.

3.5.2.1.1 Wire feeder. The wire feeder shall be a constant voltage, push type feeder which is capable of feeding soft wire in (aluminum) in 0.035, 0.045, 3/64, 1/16, and 3/32 diameters. The unit is to come equipped to feed 0.035, 3/64, and 1/16 diameter aluminum 5356 wire. Minimum characteristics shall be as follows:

(a) Wire feed speed from 50 to 600 inches per minute. The wire feed speed control shall maintain a preset feed speed within $\pm$ 2 percent throughout the entire speed range.

(b) Fast wire stopping shall be accomplished by a braking system.

(c) Drive rollers shall be designed as follows:

(1) Made of hardened steel

(2) Grooved to afford minimum wire distortion

(3) Marked for each wire size and type

(4) Capable of easy removal/replacement

(5) Capable of being aligned

(6) Required to provide smooth positive feeding over entire speed range.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:42Z
Check the source to verify that this is the current version before use.

(d) The control box shall be constructed of a strong lightweight material. It shall be insulated from electrical power and all controls, switches and connections shall be located at the front of unit.

(e) The wire spool drive shall accept standard size spools to 14 inches and shall have a cover to protect spool wire.

(f) The feeder motor shall be prelubricated, totally enclosed and capable of being braked.

3.3.2.1.2 Cables. The cables shall be designed and supplied as follows. Both of the cables in (a) and (b) can be enclosed in one cable.

(a) One 50-foot contactor cable, detachable at both ends, with polarized twist-lock connector at power source end and appropriate connector at wire feeder end.

(b) One 50-foot auxiliary power cable detachable at both ends with appropriate connectors at each end.

(c) A receptacle to receive female connector of electrode cable specified in 3.3.4.2.

3.3.2.1.3 Metal inert gas welding gun. The gun shall be a push type, 60 degree gooseneck, air-cooled unit. The handle section of the gun shall be constructed of a nigh temperature, high impact material capable of withstanding rough abuse without bending or cracking. The gun shall be rated at 300 A 100 percent duty cycle with argon shielding gas. The gun shall be capable of feeding soft wire from 0.035 inch to 3/32 inch diameter.

3.3.2.1.4 Service leads to gun. The service leads shall be flexible. Five separate services shall be provided by the service leads as follows:

(a) Welding power

(b) Gas

(c) Control

(d) Casing

(e) Liner

The liner shall be made of Teflon or similiar material.

3.3.2.1.5 Consumables. The manufacturer shall furnish one spool of electrode for each diameter of electrode specified in 3.3.2.1.1.

3.3.2.2 Output contactor. The welding set shall be equipped with an output contactor and switch for use with wire feeders when electrically cold electrode is required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:42Z
Check the source to verify that this is the current version before use.

3.3.2.3 Polarized twist-lock plugs. The welding set shall be equipped with polarized twist-lock plugs for wire feeder hookup.

3.3.2.4 Welding polarity switch. A positive acting switch or other means to change the polarity of the welding current for both constant current and constant voltage shall be provided. The device shall have its position marked for straight or reverse polarity.

3.3.2.5 Overspeed device. An overspeed protective device which will stop the engine if the maximum safe operating speed of the engine is exceeded shall be provided.

3.3.3 Standard equipment. The welding set shall be furnished with all standard equipment normally provided by the manufacturer including any special tools.

3.3.4 Accessories. The manufacturer shall furnish the following items as accessory equipment which shall be compatible with the welding set.

3.3.4.1 Storage cabinets. The welding set shall be equipped with storage cabinets of sufficient capacity to contain all componets and tools specified herein plus the addition of a cutting and welding torch outfit, an electric welding accessory kit, a welding electrode kit, and a welding rod set. Storage cabinets shall be of a low profile design and shall be mounted in a fashion not to interfere with preventive maintenance activity for the generator engine and components. Cabinets shall be top loading with the cabinet doors also being the roof of the cabinet. These doors shall overlap the sides so as to prohibit water from entering the cabinet. The cabinets shall be of sufficient strength to allow a man to stand/walk on them during maintenance to the welding set. The cabinets shall be grounded to the generator and frame. Appropriate compartmentation and restraining devices, drawers and shelves shall be provided to secure and protect fragile, small or corrosion prone components within storage cabinets. All walking surfaces shall be provided with an antiskid surface. The cabinets shall be fastened to the trailer (see Figure 1) without modification to the trailer, with the addition of fasteners only.

3.3.4.2 Welding cables. The welding set shall be furnished with one set of welding cables as follows:

(a) Ground Cable - one 50-foot length of 2/0 (34NWG), fluted, natural rubber, extremely flexible cable with one quick disconnect and one ground clamp attached.

(b) Electrode cable - one 50-foot length of 1/0 fluted, natural rubber, extremely flexible cable with one quick disconnect and electrode holder attached.

3.3.4.3 Gas cylinders. Provisions shall be made to mount one 200 cubic foot cylinder of oxygen, one 225 cubic foot cylinder of acetylene and one 330 cubic foot cylinder of argon on the welding set frame in accordance with applicable OSHA requirements. Ease of removal/replacement of cylinder is required. One empty 330 cubic foot cylinder for argon gas shall be provided with each welding set.

3.3.4.4 Regulator, pressure compressed argon gas. A two stage pressure regulator for argon gas conforming to MIL-R-19180 shall be provided with each welding set.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:42Z
Check the source to verify that this is the current version before use.

3.3.4.5 Hose assembly. A 50-foot long, black in color, hose assembly with connectors and adapters for mating one end with a gas regulator (3.3.4.4) and the other end with the wire feeder (3.3.2.1.1) in accordance with ZZ-H-461 shall be provided with each welding set.

3.3.5 Lifting and tiedown attachments. The welding set shall be equipped with lifting and tiedown attachments conforming to type III of MIL-STD-209. The welding set shall have lift points to allow the entire system, including the trailer, at maximum gross weight to be lifted as a unit by a sling.

3.3.6 Lubricants. All moving parts subject to wear, or areas requiring lubrication, shall be provided with a means of lubrication. All lubrication systems shall have simple means to add or change lubricant and shall have means for sealing all openings to prevent dirt and foreign matter from entering. Grease and oil fittings shall be easily accessible. The welding set shall be designed to operate efficiciently and satisfactorily when lubricated with standard military lubricants conforming to MIL-L-2104 and MIL-L-46167 for engine oils; MIL-L-2105 for gear oil; and MIL-G-10924 for automotive grease. Each welding set shall be serviced with military lubricants suitable for operation at 60° F.

3.3.7 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances which will permit installation of replacement parts without modification of part or machine.

3.3.8 Safety and health requirements. Covers or other safety devices shall be provided for all parts of the welding set presenting safety hazards which can be guarded without substantial interference to operation of the welding set. All guards shall provide easy access to guarded parts. The welding set shall comply with all applicable Environmental Protection Agency (EPA) and Department of Labor OSHA requirments in effect on date of manufacture.

3.4 Reliability. Each welding set produced shall perform as reliably as the first article.

3.5 Construction. The welding set shall be constructed so that when operated it can be used for any purpose for which it was designed. The welding set shall be constructed of parts which are sound in structure and free from any repair. Current state of the art technology and acceptable commercial practice shall be used in the design to minimuize corrosion, galvanic deterioration and wear points.

3.5.1 Steel fabrication. Steel used in the fabrication of the welding set shall be free from kinks and sharp bends. The straightening of material shall be done by methods which will not cause injury to the metal. Shearing and chipping shall be done neatly and accurately. Corners shall be square and true. Flame cutting, using a tip suitable for the thickness of metal, may be employed instead of shearing or sawing. Reentrant cuts shall be made in the best possible manner. All bends of a major character shall be made with controlled means in order to insure uniformity of size and shape. Precautions shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly.

3.5.2 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers, lockwashers or other positive locking shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement.

Source: https://assist.dla.mil - Downloaded: 2026-10-20T22:42Z
Check the source to verify that this is the current version before use.

3.5.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.5.4 Welding. The surface of the parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Spot, tack or intermittent welds for strength will not be permitted. Weld penetration shall provide transference of maximum design stress through the base metal juncture. Fillet welds shall be provided when necessary to reduce stress concentration. Welding shall conform to the procedures as prescribed by either the AWS or the ASME for the type of welding operation to be performed.

3.5.5 Machine work. Tolerances and gages for metal fits shall conform to the limitations specified herein and to the standards of good commercial practice.

3.5.7 Castings. Castings shall be sound and free from defects which might render the casting unsound for use.

3.5.7 Threaded parts. Threaded parts shall be in accordance with the applicable SAE fastener standard. All hex bolts and capscrews shall be SAE grade 5 or 8 and shall be so marked.

3.6. Maintainability. The welding set design, together with component and accessory location and installation, shall permit ready accessibility to all items requiring periodic maintenance service in the field. Access doors shall be capable of being held in the open position. Maintenance shall be accomplished with the use of conventional general purpose tools associated with equipment of this nature. The replacement and adjustment of components and accessories shall be accomplished with minimum drainage requirements and minimum disturbance to other elements of the welding set.

3.7. Transportability. The welding set shall be designed so that it can be transported by air, rail highway, and water transport when mounted on the M353 trailer (see figure 1). The welding generator and power unit shall be skid mounted and designed to mount on the M353 trailer, without modification to the trailer, with addition of fasteners only. The mounting and removal from the trailer shall be with a minimum of effort. The skid design shall provide the capability to lift and move the welding generator and power unit with a forklift. The dimensions of the fork tines which will be used to move the welding generator and power unit shall be as follows:

(a) Fork tines (2.5 inches thick x 8.0 inches wide)
(b) Minimum spread of fork tines 10 inches inside to inside.
(c) Maximum spread of fork tines 60 inches outside to outside.

3.8 Performance. The welding set shall provide a satisfactory power source for producing the welds specified herein, for producing auxiliary electric power specified herein, and for the following welding and related operations:

(a) Manual welding, with all types of dc covered electrodes of all sizes within the welding set's capacity, and in all positions for which the electrodes are suitable.

(b) Metal inert gas welding - semiautomatic.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:42Z
Check the source to verify that this is the current version before use.

3.8.1 Constant current operation. When adjusted for constant current operation the arc-welding power supply shall have characteristically drooping V-A curves producing relatively constant current with a limited change in load voltage. The maximum open circuit voltage shall be within the range of 75 to 100 V when the generator is set for maximum excitation and the engine is operating at a speed not exceeding the rated speed of the welding generator. The minimum open circuit voltage shall be at least 15 V less than the maximum open circuit voltage but in no case shall the minimum be more than 65 V. The current shall be adjustable over the entire range of the welding set in one or more steps with infinite stepless adjustment within each range.

3.8.2 Constant voltage operation. When adjusted for constant voltage operation, the arc welding power supply shall have characteristically a flat V-A curve, producing relatively constant voltage with a change in load current. The welding generator shall be designed and equipped with suitable controls to permit the operator to adjust the slope of the V-A curve from a relatively flat position in which the change in voltage between 100 A and maximum amperage does not exceed 10 V, to a maximum slope in which the rate of change of voltage with respect to A is three V per hundred A. The slope controls shall also provide for at least one additional intermediate setting between the minimum and maximum slope positions. The effective open circuit voltage shall be within the range of 13 at 45 V. The voltage shall be intinitely adjustable within the specified range.

3.8.3 Temperature. Without using auxiliary power from external sources, the welding set shall be equipped to start and operate under continuous operation conditions in any ambient temperature from -25°F to +125°F without exceeding the engine manufacturer's recommended limits for radiator top tank temperature and engine oil temperatures (sump and gallery). The welding set shall not be damaged when stored for 30-day periods at temperatures as low as -65°F or as high as +160°F.

3.8.4 Altitude. The welding set shall be powered with an engine which will provide the power required to perform the requirements specified herein at 5,000-foot altitude.

3.8.5 Rain and wind. The welding set shall be designed to withstand extended periods of high winds and heavy rains during operation and storage. Under these conditions, there shall be no evidence of leakage into the intake filter, exhaust system, engine crankcase or other reservoirs and the engine shall start in 2 minutes after exposure.

3.8.6 Stabililty. The welding set shall be designed to operate satisfactorily when inclined at an angle of 15 degrees in any direction from the normal operating position.

3.8.7 Vibration. The engine and generator assembly shall be designed to operate with vibration characteristcs below the level which will adversely affect the acceptable operation of the machine as specified herein.

3.8.8 Operating temperature. The welding generator shall conform to the operateing temperature requirements of NEMA EW-1.

3.9 Electromagnetic interference. The electromagnetic radiation limits of the welding set shall conform to MIL-STD-461, class III C when not welding.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:42Z
Check the source to verify that this is the current version before use.

3.10 Dimensions. The maximum outside dimensions of the welding set, including the Government furnished trailer, shall not exceed the following:

(a) 95 inches wide
(b) 96 inches high
(c) The angle of departure shall not be less than 25 degrees.

3.11 Weight. The maximum weight of the complete welding set, with all components specified herein stored and secured in appropriate locations, shall be not more than 7,000 pounds.

3.12 Cleaning, treatment and painting. The welding generator, engine, skid, cabinets and all exterior mounted components shall be cleaned, treated and painted as specified in MIL-T-704, type A. The finished coat shall be in accordance with MIL-E-52798 (forest green).

3.13 Marking. The entire welder shall be free of all attached, appended or herein.

3.13.1 Identification marking. The welder shall be marked in accordance with MIL-STD-130. A U.S. Marine Corps name plate shall be permanently attached in a conspicuous location. The name plate will be furnished, upon request, by the Commandant of the Marine Corps (Code LMB), Headquarters U.S. Marine Corps, Washington, D.C. 20380. Data applicable to blank spaces 1, 3, 11, and 12 on the name plate will be furnished by the procurring activity. The contractor shall be required to imprint the data in all blank spaces.

3.13.2 Registration Data. The registration number assigned to each welding set shall be painted on the welding set in lusterless black (FED-STD-595 No. 37038) conforming to MIL-E-52798. Registration numbers shall be applied on each side of the welding set in 1-inch characters in accordance with MIL-STD-642, except the national symbol shall not be applied.

3.13.3 Instruction plates. The welding set shall be equipped with suitably located instruction plates describing any special or important procedures to be followed in starting, operating, shutdown, and servicing the welding set or its components. Plates shall be of a durable material and shall be securely affixed.

3.13.4 Stenciling. The gross weight of each welding set shall be stenciled on each side of the welding set (with 1-inch characters)(MIL-STD-642) in lusterless black (FED-STD-595 No. 37038) in such a manner as to be readily discernible to dock and military personnel.

3.14 Workmanship. All parts, components, and assemblies of the welding set including castings, forgings, molded parts, stampings, bearings, seals, machine surfaces, and welding parts, shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. External surfaces shall be free of burrs, sharp edges, and corners, except when sharp edges or corners are required.

4. QUALITY ASSURANCE PROVISIONS.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:42Z
Check the source to verify that this is the current version before use.

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article (see 4.3)
(b) Production (see 4.4)
(c) Preparation for delivery (see 4.5)

4.3 First article. First article inspection shall be performed on the first production item. This inspection shall include the examination in 4.6 and the tests in 4.7. The first article may be a standard production item from the supplier's current inventory, provided the welding set meets the requirements of this specification and is representative of the design, construction, and manufacturing technique applicable to the remaining welding sets to be furnished under the contract.

4.4 Production. Each welding set produced in fulfillment of a Government contract or order shall be completely assembled, adjusted, lubricated, or otherwise serviced for operation. The engine shall be started and subjected to a warm-up period as recommended by the manufacturer. Each welding set shall be subjected to examination in 4.6. The welding set shall be given a run-in test and all controls shall be operated a sufficient number of times as assure that the components and mechanisms actuated by the controls operate promptly, fully, and without restriction or malfunction.

4.5 Packaging. The preservation, packaging, packing, and marking of the welding set shall be inspected to verify conformance to requirements of Section 5.

4.6 Examinations. Each welding set shall be examined for compliance with the requirements specified in Section 3 of this specification. Any redesign or modification of the supplier's standard product to comply with requirements specified herein or any necessary redesign or modification following failure to meet requirements specified herein shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any requirements specified herein or presence of one or more defects preventing or lessening efficiency shall constitute cause for rejection.

4.7 Tests. All tests shall be performed in accordance with applicable procedures of NEMA EW-1. Failure of the welding set to pass any of the following tests or failure of any component during testing shall be the basis for rejection.

4.7.1 Operational test. The welding set shall be operated for a minimum of 60 minutes during which time all components shall be individually connected and operated, safety devices identified and actuated, and all controls actuated, and if required, adjusted to insure that the welding sets operates properly. During the test, wire feed motors, welding guns, remote control devices and gas control valves shall be checked for operability. Incorrect functioning of any component shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:42Z
Check the source to verify that this is the current version before use.

4.7.2 Dimensions and weight. The dimensions and weight specified in 3.10 and 3.11 shall be verified.

4.7.3 Engine horsepower. The manufacturer shall certify that the engine manufacturer has approved the engine for application as specified here and shall furnish (1) a certified copy of the dynmometer test performed on a sample engine per SAE J816b, (2) a performance curve of fully equipped engine as installed in welding set, and (3) derating factors applicable when the engine is operated at altitudes above those specified in 3.8.4 and ambient temperatures above 85°F.

4.7.4 Performance tests.

4.7.4.1 Welding generator. The leads of the welding set shall be connected to a resistance load bank which is rated in excess of the maximum potential of the generator. The welding set shall be operated through successive periods of 10 minutes each in increments of 25, 50, 75 and 100 percent of rated output. This test shall be in addition to the test specified in 4.7.1 and shall be applied to both modes of performance as specified in 3.8.1 and 3.8.2.

4.7.4.2 Wire feed mechanisms. This test shall be applicable to the wire feeder/gun and cable assembly specified in 3.3.2.1. The wire shall be fed through the gun and cable assembly with all of the cable assembly coiled to 24 inch diameter inside diameter. The direction of wind of the coil shall be in the opposite direction of the wind of the electrode wire on the reel or wire holder. The wire feed drive shall be energized and there shall be no evidence of the wire feed motor being overloaded while feeding the wire through the conduit at the minimum and maximum feed rates specified in 3.3.2.1.1. The test shall be conducted using the smallest and largest sizes of soft wires specified in 3.3.2.1.1.

4.7.5 Welding tests. Actual welding operations shall be performed during the first article tests. The welding set shall be adjusted in accordance with the manufacturer's recommendation. Welds not less than 12 inches long shall be performed by the subassembly unit specified in 3.3.2.1. The operator shall also perform manual welding utilizing the cables specified in 3.3.4.2 and 1/8 and 1/4 inch diameter electrodes applicable for the processes. The operator shall deposit 12-inch-long weld beads using dc with both polarities. There shall be no evidence of difficulty in striking or maintaining the arc.

4.7.6 Lifting and tiedown attachments test. The welding set shall be tested to verify that the attachments conform to the requirements specified in 3.3.5. Failure of any component, or breakage, distortion or deformation shall constitute failure of this test.

4.7.7 Wind and rain test. The wind and rain test shall be conducted as follows:

(a) TEST APPARATUS:

(1) Timing device.

(2) Measuring device (U.S. Weather Bureau type preferred).

(3) Spray fixture.

(b) TEST PROCEDURE:

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:42Z
Check the source to verify that this is the current version before use.

(1) Subject the welding set to a simulated rainfall of not less than 4 inches per hour. The simulated rainfall shall impinge on the welding set at an angle of 45 degrees, ±5 degrees, with a force equivalent to a 30 mile per hour wind. The simulated rainfall shall disperse uniformly over the surfaces of the welding set.

(2) Expose the top and left side of the welding set to the rain described above for a period of 5 minutes with engine off. With rain still flowing, start engine and allow engine to idle for 2 minutes.

(3) Repeat (b) with the top and front exposed.

(4) Repeat (b) with the top and right exposed.

(5) Repeat (b) with the top and rear exposed.

(6) Stop the rainfall and allow the welding set to stand for not less than 30 minutes with the engine off. Drain a small quantity of fluid from each fluid reservior (except cooling system) and examine for presence of water.

(7) Check electrical components, gages, cab (if furnished), and the air cleaner for water entry and damage.

4.7.8 Test for electromagnetic interference characteristics. The welding set shall be tested in accordance with MIL-STD-462 for class III equipment. Inability to pass the test shall constitute failure of this test.

5. PACKAGING

5.1 Preservation-packaging. The preservation-packaging shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. The complete welding set shall be preserved and packaged in accordance with level A of MIL-W-45562.

5.1.2 Commercial. The complete welding set shall be preserved and packaged in accordance with the contractor's standard practice.

5.2 Packing. The packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. The complete welding set shall be packed in accordance with level A of MIL-W-45562.

5.2.2 Level B. The complete welding set shall be packed in accordance with level B of MIL-W-45562.

5.2.3 Commercial. The complete welding set shall be packed in a manner which will insure arrival at destination in satisfactory condition and which will be acceptable to the carrier at lowest rates. Packing shall comply with Uniform Freight Classification Rules, or National Motor Freight Classification Rules.

5.3. Marking. The welding sets, attachments, interior packages, and shipping containers shall be marked in accordance with MIL-STD-129.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:42Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. The welding set covered by this specification is to provide a dc power source where other electric power is not available; where portability is desired in shops; field maintenance and construction; and where ac auxiliary electric power is desired in connection with the welding operation. The welding set described herein is intended for domestic and overseas military operations. The welding set shall be subject to the rigors of amphibious loading and unloading from naval vessels, as well as sea transport and other conditions inherent to military operations.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.

(b) Level of preservation, packaging and packing (5.1 and 5.2).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) the requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data requirements	Applicable DID no.	Options
4.3	First Article Inspection Procedure	DI-T-4901	-
4.3	First Article Inspection Report	DI-T-4902	-

(Copies of DID's required by contractors in connection with specific asquisitio functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.

6.3 First article. Any changes or deviations of production welding sets from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the suppli of his obligation to furnish welding sets conforming to this specification.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - MC
Project No. 3431-N152

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:42Z
Check the source to verify that this is the current version before use.

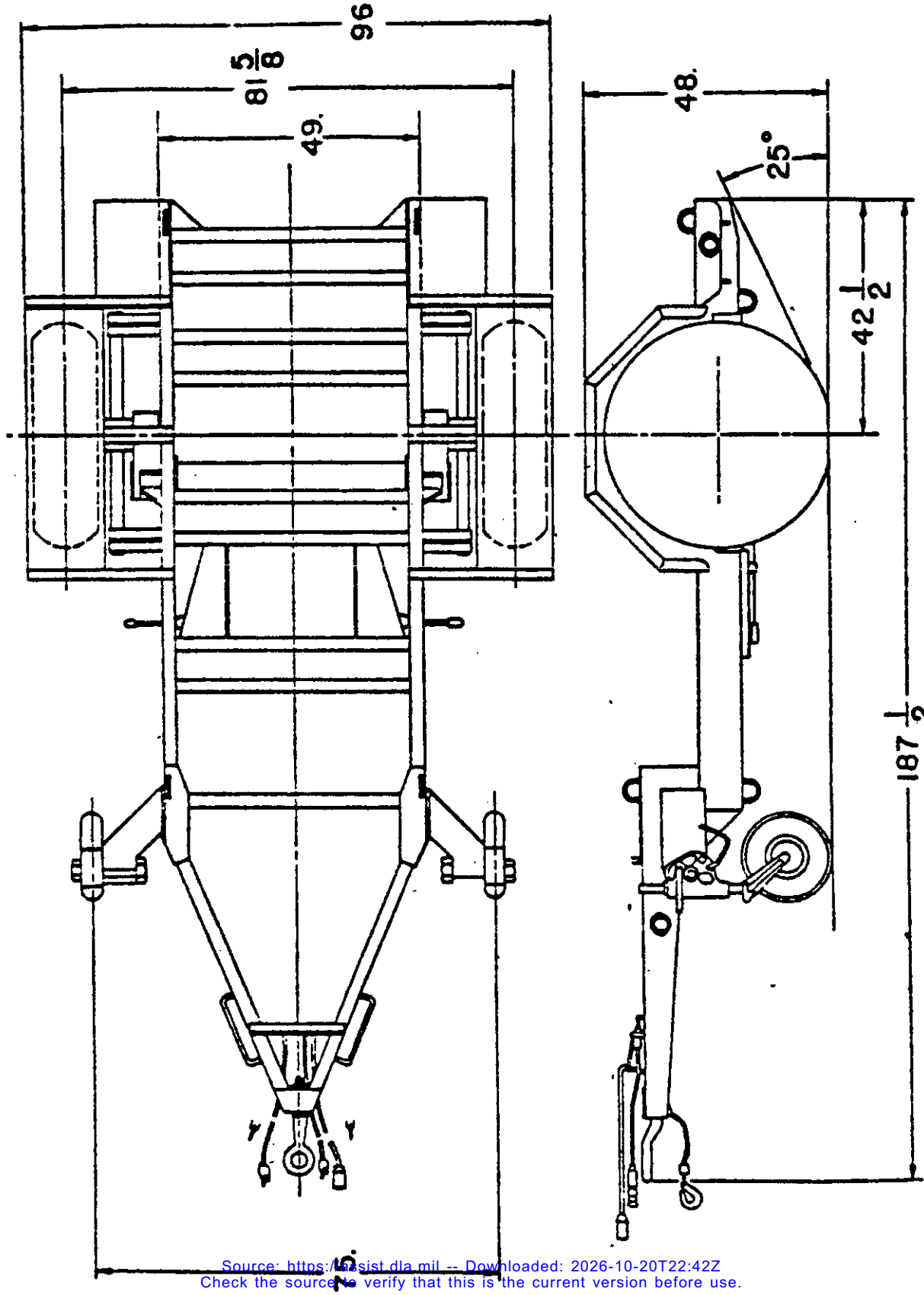


Figure 1. M353 Trailer

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:42Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
Headquarters, U.S. Marine Corps
Washington, D.C. 20380

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

POSTAGE AND FEES PAID



Commandant of the Marine Corps
Headquarters, U.S. Marine Corps
(Code LMA-1)
Washington, D.C. 20380

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:42Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE MIL-W-29383A(MC)
WELDING SET, ARC: AC/DC, MOBILE MULTIPROCESS, DIESEL ENGINE DRIVEN

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
☐ YES ☐ NO (If "Yes", in what way?)

4. REMARKS

SUBMITTED BY (Printed or typed name and address - Optional)

TELEPHONE NO.

DATE

DD FORM 1426 PREVIOUS EDITION OF 1 JAN 72 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:42Z
Check the source to verify that this is the current version before use.